MULTI-CLOUD AI AGENT DESIGN SPECIFICATION

CipherMesh

Multi-Cloud LLM Privacy and Secret-Protection Agent

MISSION Prevent personal data, cloud credentials, proprietary infrastructure details, and other secrets from leaving an application through an LLM prompt or response—regardless of whether the destination is AWS Bedrock, Azure AI, GCP Vertex AI, or a local model.

Field	Submission detail
Student	Kumar Waibhav Akshat
Register No.	RA2212704010027
Programme	M.Tech Integrated - CSE Data Science
Section	AG1
Course	21CSE525T - Agent Technology
Assignment	1.1 - Project Ideation, Environment Modeling, and PEAS Matrix Definition

1. Research-Backed Domain and Agent Persona

1.1 Existing real-world problem

Developers and customers paste source code, stack traces, configuration fragments, support conversations, documents, and operational logs into LLM applications. These inputs may contain cloud credentials, personal information, internal endpoints, or proprietary data. If the application sends the prompt first and checks later, the disclosure has already crossed the trust boundary. The risk is provider-independent: the same application may route requests to Amazon Bedrock, Azure AI, Google Vertex AI, or a local model.

This is an established industry problem. OWASP lists Sensitive Information Disclosure among the 2025 Top 10 risks for LLM applications. AWS Bedrock Guardrails can block or mask sensitive prompt data; Microsoft recommends input filtering and monitoring for AI systems; Google warns that exposed service-account private keys can enable authentication as the service account. The convergence of these controls establishes a clear need for protection before inference.

1.2 Proposed agent persona

CipherMesh is an autonomous privacy gateway positioned between users/applications and model providers. For every request and response it perceives content and routing context, detects sensitive entities, determines risk, and independently selects one of four actions: ALLOW, REDACT, REVIEW, or BLOCK. It stores only privacy-safe fingerprints and finding metadata in its audit trail. The Assignment 1.2 implementation is a transparent simple reflex agent; the production path adds contextual NLP, policies, adapters, memory, approvals, and authenticated deployment.

1.3 Stakeholders, boundary, and impact

- Developers: avoid accidentally sending credentials, internal URLs, customer records, or code secrets to an external model.
- Customers: receive real-time protection when entering personal data into AI chatbots and support assistants.
- Security/privacy teams: gain consistent provider-neutral policy enforcement and auditable decisions.
- Boundary: CipherMesh inspects application-layer text before provider transmission and after model response; it does not claim to replace IAM, secret managers, encryption, contractual controls, or provider-native guardrails.
- Impact measures: blocked credential disclosures, correctly redacted PII, low false-block rate, inspection latency, coverage, and privacy-safe audit completeness.

2. Zero-Cost Build and Showcase Feasibility

Component	Free implementation	Purpose
Agent/API	Python standard library; later FastAPI	Real-time request/response interception
Detection	Regex/checksum rules; Microsoft Presidio	Secrets and contextual PII recognition
Local AI	Ollama with an open-weight model (optional)	Contextual classification without inference fees
Storage	SQLite or append-only JSON audit events	Local privacy-safe evidence
Interface	Local web UI or CLI	Live classroom demonstration
Packaging	Docker/Compose; optional local Kubernetes	Reproducible production-style deployment
Cloud adapters	Provider SDK-compatible interfaces and mock endpoints	Show AWS/Azure/GCP routing without paid calls

COST CLAIM CipherMesh can be built and demonstrated locally with zero software or API cost. Paid model calls and permanent public hosting are optional and excluded from this claim.

3. PEAS Profile

PEAS element	CipherMesh specification
Performance measures	High secret/PII recall and precision; zero critical-secret forwarding in known cases; low false-block rate; low inspection latency; correct ALLOW/REDACT/REVIEW/BLOCK decision; provider availability; privacy-safe logging; policy traceability.
Environment	Developer tools, customer chat applications, API gateway, AWS Bedrock, Azure AI, GCP Vertex AI, local Ollama, prompts, responses, source/log/config text, privacy policies, approval users, audit store, and adversarial or accidental inputs.
Actuators	Allow unchanged; redact/mask; tokenize; block transmission; hold for approval; warn developer/customer; select local-only route; generate safe preview; create privacy-safe audit event; recommend credential rotation.
Sensors	Prompt/response text; provider and model destination; request ID/time; pattern and NLP findings; credential type; PII type; confidence; content length; policy version; user/tenant context; approval and downstream result.

4. Task Environment Classification

Dimension	Classification	Justification
Observability	Partially observable	The agent sees submitted content and metadata, but not every external data source, provider process, or user intention.
Determinism	Stochastic	Natural language is ambiguous; contextual detectors and LLM classifiers may return uncertain or incorrect results.
Episodicity	Sequential	Earlier decisions affect user trust, audit state, repeated requests, approvals, and future policy behavior.
Dynamics	Dynamic	Requests arrive continuously; providers, policies, credential formats, and attack techniques change.
State/action space	Mixed	Text and confidence are high-dimensional/continuous; provider, finding, and action types are discrete.
Agents	Multi-agent	CipherMesh interacts with users, applications, model APIs, approval services, security agents, and local/cloud models.
Knowledge	Partly known	Known formats and policies are explicit; novel secrets, context, and user intent may be unknown.

5. Perceive-Reason-Act Loop

Stage	Assignment 1.2	Production evolution
Perceive	Read one JSON LLM request.	Intercept authenticated request/response streams.
Parse	Validate provider, model, ID, size, and text.	Verify schema, identity, tenant, signature, freshness, and quota.
Reason	Evaluate deterministic detection rules.	Combine rules, Presidio, policy, context, and optional local LLM.

Stage	Assignment 1.2	Production evolution
Act	ALLOW, REDACT, REVIEW, or BLOCK.	Forward through provider adapter or approval workflow.
Audit	Log hash, finding type, rule, and safe preview.	Write encrypted/immutable metadata without raw secrets.
Repeat	Process the next sensory event.	Scale concurrent workers behind a durable queue.

6. Condition-Action Policy

Rule	Current percept condition	Action
R1	AWS access key, GCP private-key header, Azure SAS signature, or generic API token	BLOCK; never transmit; show safe preview; recommend revoke/rotate
R2	Email, Indian phone number, or IP address without a critical secret	REDACT; forward only sanitized text
R3	Internal/corporate URL without a critical secret	REVIEW; hold for explicit approval
R4	No sensitive finding	ALLOW under selected provider adapter
R5	Malformed envelope, unsupported provider, empty content, or oversize prompt	REJECT envelope safely and continue
Priority	More than one rule matches	Highest risk wins: BLOCK > REVIEW > REDACT > ALLOW

The included examples are synthetic and intentionally invalid. Production detection must be evaluated against labeled datasets, tuned for context, and layered with provider-native controls because automated detectors cannot guarantee that every sensitive item will be found.

7. Autonomous Agent vs. Standard Object-Oriented Program

Aspect	CipherMesh autonomous agent	Standard OOP program/object
Control	Owns a continuous perceive-decide-act loop.	Waits for an external caller to invoke methods.
Situated behavior	Operates at a live trust boundary between application and models.	Represents data/functionality without inherent environmental placement.
Decision authority	Chooses ALLOW, REDACT, REVIEW, or BLOCK from policy and percepts.	Executes the operation selected by surrounding control flow.
Initiative	Stops unsafe transmission and warns the user without a separate command.	Does not initiate unless orchestrated externally.
State	Percepts, findings, policy version, obligations, approvals, and audit context.	Data fields and lifecycle state.
Communication	Exchanges typed messages with apps, providers, humans, and other agents.	Calls known interfaces or returns values.
Adaptation	Later updates policies and plans from feedback and new detector knowledge.	Changes only through explicitly coded application logic.
Example	Detects a cloud key inside a prompt and blocks it autonomously.	A scanner object returns matches; another component must decide what happens.

8. Production-Ready Architecture and Course Roadmap

Layer	Design
Gateway	OpenAI-compatible local HTTP proxy with provider adapters for Bedrock, Azure AI, Vertex AI, and Ollama.
Detection	Fast deterministic secret detectors, Presidio PII analyzer/anonymizer, optional local classifier, and versioned policies.
Privacy	Never persist raw credentials; hash request content; encrypt configuration; minimize logs; configurable retention.
Security	Authentication, tenant isolation, least privilege, TLS, rate limiting, signed policy bundles, deny-by-default provider routing.
Reliability	Timeouts, idempotency keys, bounded queues, retry policy, health probes, circuit breakers, metrics, and dead-letter handling.
Quality	Unit/integration/security tests, labeled detector evaluation, false-positive review, latency benchmark, and red-team corpus.

Course unit	CipherMesh extension
Unit 2	Concurrent scanning workers, daemon mode, sockets/RPC, and typed provider messages.
Unit 3	Privacy, security, routing, and approval agents coordinate; negotiate provider selection under policy.
Unit 4	Knowledge representation, adaptive policies, BDI plans, Presidio/local-LLM integration, and provider APIs.
Unit 5	Authenticated requests, signed policies, hostile prompt handling, host integrity, audit controls, and secure deployment.

9. Rubric Compliance Checklist

Requirement	Evidence	Status
Domain & Persona Clarity (1)	Existing disclosure problem, users, trust boundary, impact, and persona in Section 1.	Complete
PEAS Profile Completeness (2)	Detailed performance, environment, actuators, and sensors in Section 3.	Complete
Agent vs. Object Contrast (2)	Eight structural comparisons grounded in Unit-1 paradigms in Section 7.	Complete
Environment classification	Seven formal dimensions with justification in Section 4.	Complete
Free real-time feasibility	Concrete zero-cost stack and honest cost boundary in Section 2.	Complete

10. Research Sources

- OWASP Top 10 for LLM Applications 2025, LLM02 Sensitive Information Disclosure. https://owasp.org/www-project-top-10-for-large-language-model-applications/assets/PDF/OWASP-Top-10-for-LLMs-v2025.pdf
- Amazon Bedrock Guardrails, sensitive-information filters: detect, block, or mask PII in prompts and responses. https://docs.aws.amazon.com/bedrock/latest/userguide/guardrails-sensitive-filters.html
- AWS IAM, access-key management: access keys are long-term credentials and must not be placed in application files. https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_access-keys.html
- Microsoft Azure AI security best practices: input/output filtering, prompt-injection protection, API controls, and monitoring. https://learn.microsoft.com/en-us/azure/security/fundamentals/ai-security-best-practices

- Microsoft Azure Storage security: SAS signature portions should be redacted when shared for troubleshooting. https://learn.microsoft.com/en-us/azure/storage/common/storage-explorer-security
- Google Cloud IAM, service-account key management: leaked private keys enable unauthorized authentication and should not enter repositories. https://docs.cloud.google.com/iam/docs/best-practices-for-managing-service-account-keys
- Microsoft Presidio: open-source PII detection, redaction, masking, and anonymization for Python, Docker, and Kubernetes. https://github.com/microsoft/presidio
- Ollama API documentation: local model API for a zero-inference-fee demonstration. https://docs.ollama.com/api/introduction
- 21CSE525T syllabus, assignment plan, course handout, and Unit-1 lecture decks supplied for this assignment.